

AI-Driven Tea Quality and Production Improvements in Mid-Country Sri Lanka

Project ID: 25-26J-133

Project Proposal Report

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DECLARATION

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Tea production is a key part of Sri Lanka's economy, especially in the mid-country region known for its premium varieties. However, diseases, pests, and invasive species affecting tea plantations reduce yield and quality, causing significant economic losses for growers. Traditional detection methods rely on manual inspections, which are slow, labor-intensive, and often subjective.

This study presents an AI-driven system for early detection of tea leaf diseases, pest infestations, and invasive species, along with practical treatment guidance. Using image-based deep learning with Convolutional Neural Networks (CNNs), the system can identify diseases such as blister blight and red rust, detect pests, and recognize invasive species like *Lantana camara*, *Tridax procumbens* etc. High-resolution images collected from plantations will be annotated with expert input, and advanced models like R-CNN may be used for precise localization of affected areas.

Once a problem is detected, a rule-based engine recommends effective treatment strategies, including chemical, biological, and organic options. With expected detection accuracy above 90%, the system enables timely interventions, reduces pesticide misuse, and improves tea quality. The solution will be delivered as a mobile decision-support tool, helping small-scale estate holders quickly identify and manage problems, promoting sustainable practices and enhancing the global competitiveness of Sri Lankan tea.

Keywords: Tea Leaf Disease Detection, Pest Infestation Detection, Treatment Recommendation System, Artificial Intelligence (AI), Deep Learning, Convolutional Neural Networks (CNN), Invasive Species

Table of Contents

DECLARATION	i
ABSTRACT	ii
LIST OF FIGURES	iv
LIST OF TABLES	v
LIST OF ABBREVIATIONS	vi
1. INTRODUCTION	1
2. Background & Literature survey	1
2.1 Research Gap	7
2.2 Research Problem	10
3. OBJECTIVES	12
3.1 Main Objectives	12
3.2 Specific Objectives	12
4. METHODOLOGY	14
4.1 Project Overview	14
4.2 System Architecture Diagram	15
4.3 System Overview	16
4.4 Software Development Process	16
4.5 Requirements Gathering and Analysis	17
4.5.1 Feasibility Study	21
4.5.2 Dataset	21
4.5.3 Implementation	22
4.5.4 Testing	22
4.5.5 Grant Chart	22
4.5.6 Work Breakdown Structure	22
5. PROJECT REQUIREMENTS	22
5.1 Functional requirements	22
5.2 Non-functional requirements	22
5.3 Software requirements	23
5.4 User requirements	23
5.5 Expected Test Cases	23
6. COMMERCIALIZATION	25
7. BUDGET AND BUGET JUSTIFICATION	26
REFERENCES	27
APPENDICES	30

LIST OF FIGURES

Figure 1. 1 : Tea plantation areas in Sri Lanka	7
Figure 1. 2 : Diseases of Tea Plant.....	8
Figure 1. 3 : Tea Leaf Diseases	9
Figure 1. 4 : Invasive Species.....	9
Figure 1. 5 : Sample methodology for predicting the tea leaf diseases.....	10
Figure 1. 6 : Examples of Data Augmentation.....	10
Figure 1. 7 : Sample interface of the recognition process and results of the model	11
Figure 3. 1 : System Architecture.....	14
Figure 3. 2 : System Architecture of disease detecting component.....	16
Figure 3. 3 : Agile Methodology	16
Figure 3. 4 : Grantt chart	17
Figure 3. 5 : Work Breakdown Structure	19

LIST OF TABLES

Table 1. 1: Comparative Analysis of Existing Researches	15
Table 3. 1 : Technologies, techniques, architectures and algorithms.....	23
Table 4. 1 : Expected Test Case1	39
Table 4. 2 : Expected Test Case2	40
Table 6. 1 : Expenses for the proposed system	35

LIST OF ABBREVIATIONS

Abbreviation	Description
CNN	Convolutional Neural Networks
TRI	Tea Research Institute
ML	Machine Learning
AI	Artificial Intelligence
SVM	Support Vector Machine
UAT	User Acceptance Testing
MLP	Mutlilayer Perceptron
RCNN	Region-Based Convolutional Neural Network
YOLO	You Only Look Once
WBS	Work Breakdown Structure
SCRUM	Agile project management framework

1. INTRODUCTION

1.1 Background & Literature survey

Tea cultivation plays a crucial role in Sri Lanka's economy, contributing significantly to exports and supporting the livelihoods of many rural communities. However, the industry faces several challenges, including crop diseases, pest infestations, and inconsistencies in leaf quality, which can impact both productivity and profitability. Traditional methods of detecting diseases are often labor-intensive, time-consuming, and prone to human error, making them less suitable for large-scale monitoring. With the rise of Artificial Intelligence (AI) and Computer Vision, modern approaches now offer accurate, timely, and cost-effective solutions to support sustainable tea production.

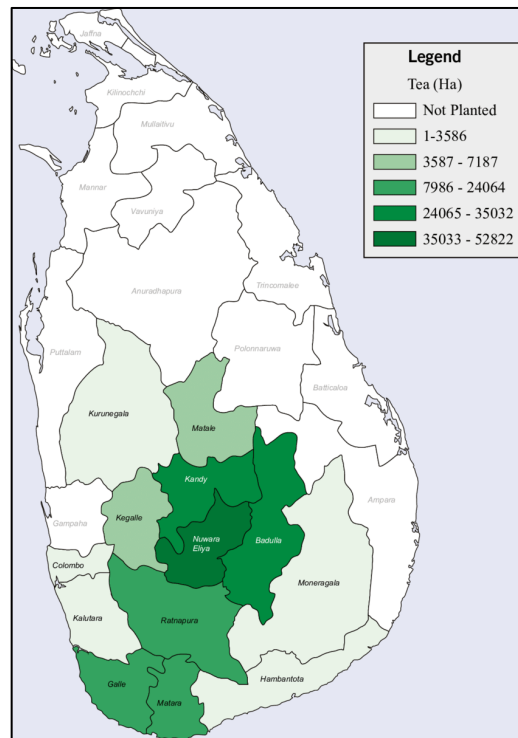


Figure 1.1 :Tea Plantation Areas in Sri Lanka

Recent studies (2023) have introduced GreenGuard, a CNN-based system for the early detection of tea leaf diseases [1]. This system demonstrated the potential of AI by accurately identifying diseases at an early stage, allowing prompt intervention to reduce crop losses. Following this, other research applied transfer learning and CNNs with the Kaggle Tea Illness Dataset to classify seven tea leaf diseases, achieving 92% accuracy, which was further improved to 95.9% using NASNet [2]. These studies highlight the effectiveness of deep learning for scalable applications and point toward the potential for combining detection with treatment guidance.

To overcome challenges posed by small sample sizes and natural field conditions, the multi-CNN model MergeModel was proposed [3]. It integrates U-Net for leaf segmentation, SinGAN-based data augmentation, and multiple CNN modules to capture both global and fine-grained features. This model achieved high accuracy in detecting diseases such as tea white scab, tea leaf blight, tea red scab, and tea sooty mould. Similarly, rapid detection of Black Rot and Rust using a lightweight CNN achieved approximately 95.93% precision, demonstrating fast and reliable disease identification while reducing dependency on chemical pesticides [4].

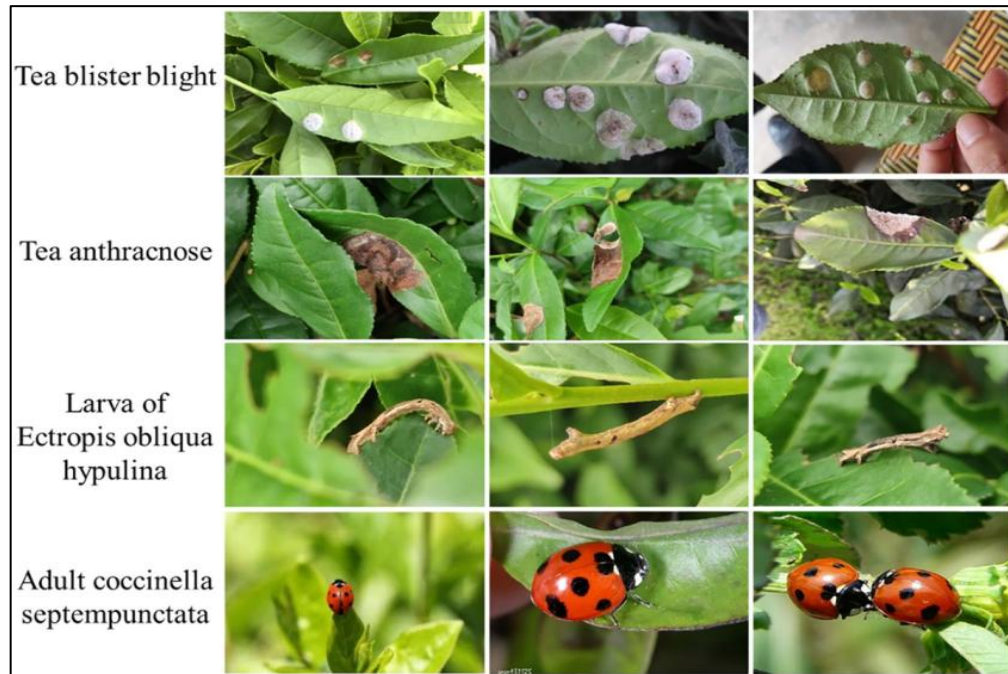


Figure 1. 2 : Diseases of Tea Plant

In India, a multi-layered CNN was used to classify anthracnose, brown blight, and red leaf spot with different optimizers, where Adam produced the highest accuracy of 83% [5]. Another improved deep CNN model reached 73% average accuracy for small-sample disease detection, emphasizing the importance of early identification in minimizing yield losses and environmental impacts [6]. The LeafNet model further highlighted detection of seven major tea leaf diseases and compared CNN with traditional machine learning models (SVM, MLP) for disease prediction and pesticide recommendation, leveraging architectures such as Inception-V3, GoogLeNet, and ResNet-110[7].

Federated learning combined with CNNs has also been explored for tea leaf disease classification, allowing multiple clients to contribute data while maintaining privacy [8]. By aggregating insights from several sources, the system classifies diseases into four severity levels mild, moderate, severe, and critical with macro-average precision, recall, F1-score, and accuracy up to 93.24%. This method not only ensures scalability but also maintains data privacy, reduces the need for centralized storage, and provides robust early detection and treatment planning.



Figure 1.3 :Tea Leaf Diseases

Advanced deep learning models, including ResNet50, ResNet101-V2, and Modified-ResNet50-V2 (M-RN50V2), have also been applied for tea leaf disease detection, with M-RN50V2 achieving the highest accuracy of 90.54% [9]. Similarly, transfer learning using large Kaggle datasets achieved 99.86% accuracy over 20 training epochs [10]. These approaches facilitate faster and more accurate identification of diseased leaves, enabling timely intervention, efficient pesticide use, and lower environmental impact.

Beyond disease detection, automated quality assessment systems such as tAssessee use CNN and Mask-RCNN architectures to evaluate disease, pest attack, freshness, and grading in both single and bulk leaves [11]. Experiments using Sri Lankan factory images showed 87% - 98% accuracy for single leaves and 91% - 100% for bulk leaves, demonstrating AI's potential to enhance quality control and operational efficiency.

Object detection algorithms have also been employed for pest and disease identification. An improved YOLOv5 model incorporating Multidimensional Collaborative Attention (MCA) and Structured Intersection over Union (SIoU) increased precision to 96.6%, recall to 95.8%, and mAP to 69.8% [12]. Earlier work using CNN-based pest identification, K-Means optimized anchor frames, YOLO-based bud detection, and Faster R-CNN with Retinex enhancement has laid a strong foundation for integrating advanced detection techniques in tea cultivation [1,2,3,4,5,6,7,12].

In addition to diseases and pests, the spread of invasive plant species poses an emerging threat to Sri Lankan tea plantations. These species compete aggressively with tea bushes for nutrients, sunlight, and water, leading to reduced yields and increased management costs. Recent studies confirm the rapid spread and ecological impact of *Lantana camara* across Sri Lanka's agricultural regions under changing climate conditions [13], as well as its negative influence on native vegetation diversity and structure [13]. Furthermore, weed management has been identified as the second-highest production cost in the Sri Lankan tea industry, with invasive weeds contributing to yield losses of up to 20% and developing herbicide tolerance [14]. These findings underscore the importance of integrating AI-based invasive species detection into tea management systems, complementing disease and pest identification to promote sustainable and cost-efficient cultivation.



Figure 1.4: Invasive species

Tea cultivation in India and Sri Lanka continues to be affected by multiple fungal and algal pathogens, including blight, dieback, charcoal stem rot, root rot, and black rot [15]. With growing awareness of the negative effects of synthetic fungicides, researchers have explored beneficial microbes such as *Trichoderma*, *Bacillus*, *Pseudomonas*, and *Actinomycetes* as sustainable alternatives for disease management. The transition to organic cultivation in Darjeeling demonstrates that microbial biocontrol can be successfully integrated with AI-based detection to improve both quality and sustainability [15].

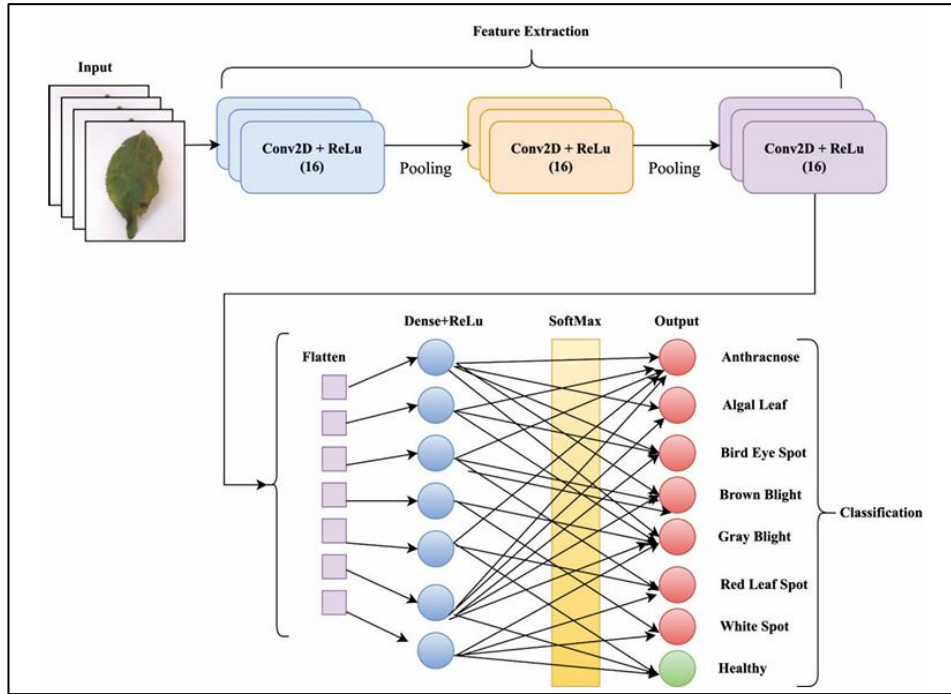


Figure 1.5: Sample Methodology for Predicting the Tea Leaf Diseases

Hybrid AI approaches combining CNNs with Random Forest classifiers have achieved over 91% precision, recall, and F1-score across eight disease categories, with an overall accuracy of 98% [16]. This integration provides a robust framework for automated disease detection, improving productivity, sustainability, and efficiency. Likewise, MobileNetV3-CA with Coordinate Attention modules has been applied to detect 17 types of tea leaf diseases and insect infestations, achieving 95.88% accuracy and 94.12% F1-score, outperforming classical models and demonstrating strong practical applicability [17].

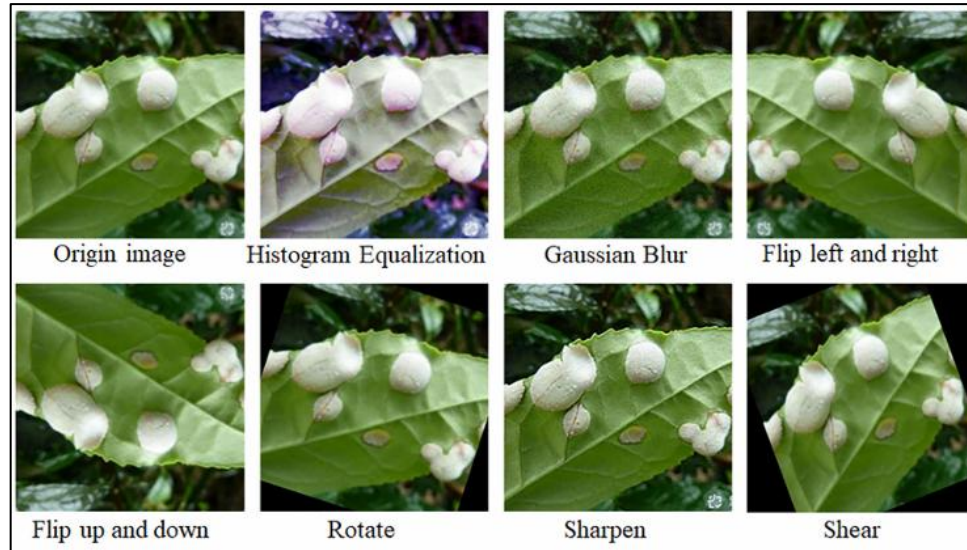


Figure 1.6: Examples of Data Augmentation

In Sri Lanka, deep transfer learning using pre-trained CNN architectures (Xception, DenseNet201, VGG16, InceptionV3, EfficientNetB0, MobileNetV2) was applied to detect six tea leaf conditions from 607 images collected in central tea gardens [18]. With fine-tuning and data augmentation, the model achieved an average accuracy of 99.58%, showcasing the ability of transfer learning to handle limited samples and support scalable, efficient, and accurate disease detection.

Collectively, these studies highlight the potential of AI and deep learning to provide accurate, scalable solutions for detecting tea leaf diseases and pests while delivering actionable treatment guidance. Building on this work, the current research leverages datasets from the Tea Research Institute (TRI) of Sri Lanka to develop an integrated framework for early disease detection, pest identification, and treatment recommendations. By using locally sourced data, the system aims to produce highly relevant and precise results tailored to Sri Lankan tea cultivation, supporting sustainable and efficient production.



Figure 1.7 : The sample interface of the recognition process, results of the model

1.2 Research Gap

Despite numerous AI-driven approaches for tea leaf disease and pest detection, significant gaps remain in existing research. Most current systems focus on detecting either diseases or pests, often relying on limited or publicly available datasets, and do not provide an integrated, real-time solution capable of handling multiple types of diseases and pests simultaneously. Furthermore, very few studies utilize locally sourced, high-resolution datasets from tea estates in Sri Lanka, which are essential for accurate, context-specific detection and actionable recommendations [1,2,3,4,5,8].

In addition to diseases and pests, the rapid spread of invasive plant species has emerged as a major but underexplored challenge in tea cultivation. Existing AI frameworks rarely address the detection or management of these invasive species, despite their proven adverse effects on yield, soil health, and biodiversity [13,14]. The absence of a comprehensive AI-based system capable of simultaneously identifying diseases, pests, and invasive species underscores a critical research gap and the need for an integrated, intelligent tea plantation management solution.

Key Research Gaps:

1. Integration of Disease and Pest Detection:

Existing systems often specialize in either disease detection or pest identification. There is a need for an integrated system that can simultaneously detect multiple major tea leaf diseases such as blister blight, red rust, brown blight and key pest infestations, including shot hole borer [1,12,15].

2. Use of Real-World Local Data:

Many studies rely on generic datasets, which may not capture the unique environmental conditions, leaf morphology, and disease patterns of Sri Lankan tea estates [3,5,16]. Leveraging real-world data from the Tea Research Institute (TRI) of Sri Lanka ensures higher accuracy and practical relevance for local cultivators.

3. High-Resolution Image Analysis:

While previous models often use low-resolution images or preprocessed datasets, high-resolution imaging is crucial for detecting early-stage infections, subtle symptoms, and small pests [3,15]. Current systems rarely combine high-resolution imaging with robust AI models for precise identification.

4. Actionable Treatment Guidance:

Current models primarily focus on classification without offering actionable recommendations. Integrating detection with treatment advice for example, pest control measures or disease management strategies can greatly enhance practical utility for tea farmers.

5. Real-Time Detection and Early Intervention:

Most current systems analyze images offline or with delays, which limits their usefulness for farmers in the field. There is a need for systems that can detect affected leaves as soon as symptoms appear, allowing for timely actions to prevent disease or pest spread and minimize crop losses [1,11].

6. Invasive Species Identification and Management:

Invasive plant species such as *Lantana camara*, *Tridax procumbens* and *Vernonia cinerea* pose a growing threat to tea plantations by competing with tea bushes for nutrients, sunlight, and water. However, very few AI-driven systems address the detection or management of these invasive species, despite their significant ecological and economic impact on tea cultivation [13,14]. Integrating invasive species detection into existing disease and pest identification frameworks would provide a more comprehensive, sustainable approach to tea plantation management.

7. Scalable and Field-Friendly Models:

Many AI models demand heavy computational resources, making them difficult to use directly in tea estates. There is a clear need for lightweight, efficient models that can run smoothly on practical devices, ensuring that large-scale, real-world deployment is feasible for farmers.

Feature	Existing Systems	Proposed System
Detection Scope	Most systems focus on a single data type, such as leaf images or pest signs, ignoring other important factors like environmental factors.	Integrates high-resolution leaf images, pest signs, invasive species images and environmental data to provide a comprehensive analysis for accurate disease, pest and invasive species identification.
Dataset	Public datasets like Kaggle, often limited in local relevance. Because of that, may not capture local disease patterns or unique environmental conditions.	High-resolution, real-world images from Tea Research Institute (TRI) and Sri Lankan tea estates
Image Resolution	Often low or preprocessed images. Because of that, cannot detect early-stage symptoms	High-resolution imaging for early and subtle symptom detection
Actionable Guidance	Few or no recommendations. Limited practical utility for farmers.	Integrated treatment and pest management guidance
Real-Time Analysis	Mostly offline or delayed detection. Not suitable for immediate field intervention	Early and precise real-time detection of affected leaves

Table 1. 1- Comparative Analysis of Existing Research

1.3 Research Problem

Sri Lanka's tea industry, a key part of the country's economy and well-known around the world, is facing a serious challenge that threatens its long-term competitiveness. Tea production has dropped to a 26-year low, mainly due to the aging of tea plants. Most plants are now over 50 years old, and there has been little replanting due to financial and logistical issues.

Smallholder farmers manage about 75% of tea lands and produce 61% of the country's tea. However, they have limited access to modern farming practices, support services, and credit. Their farming techniques are often inefficient. High labor costs, along with a long-term labor shortage, further reduce their productivity.

The sector has been hurt by sudden government policy changes, such as the 2021 ban on chemical fertilizers, which resulted in an estimated 40% loss in yields. Additionally, the ongoing economic crisis has led to fuel shortages, power outages, and disruptions in tea processing and transportation. These factors have significantly impacted Sri Lanka's position in the global tea market.

Pests traditionally lower the productivity of crops, making crop protection essential for meeting consumer demands and ensuring profitability in agriculture. The productivity of crops like tea is under threat from both pest damage and competition for resources.

In addition to these challenges, the spread of invasive plant species such as *Lantana camara*, *Tridax procumbens*, and *Vernonia cinerea* has become a growing concern in Sri Lankan tea plantations. These species aggressively compete with tea bushes for nutrients, sunlight, and water, reducing overall yield and increasing management costs. Their rapid spread, herbicide tolerance, and ecological impact further threaten the long-term sustainability of tea cultivation. However, current systems rarely include invasive species detection or control within their monitoring frameworks, leaving a critical gap in

integrated tea plantation management.

Sri Lankan smallholder tea farmers have limited access to modern technology, specialist knowledge, and effective tools for managing tea plant health. Even when such technologies are available, they are often too expensive or inaccessible. This limits farmers' ability to diagnose and treat plant diseases and pest infestations early. Without timely and accurate diagnosis, smallholders face higher crop losses and reduced product quality.

This research draws on a large, annotated dataset from the Sri Lankan Tea Research Institute that reflects the real conditions in Sri Lanka. Using this locally relevant data will improve the accuracy and reliability of the AI models, addressing a significant gap in current research that often relies on generic datasets. This makes the system very suitable for use in Sri Lanka's tea estates.

To address this issue, my research proposes an AI-based system for disease and pest detection for smallholder farmers. In this user-friendly system, farmers can easily upload images of infested or infected tea leaves using a mobile device. The AI model then identifies the disease, pest, or leaf condition, provides the scientific name along with a short description, and offers practical, eco-friendly treatment options.

This solution provides smallholders with an easy-to-use, low-cost method for early detection and effective management of tea plant health problems. It aims to reduce reliance on expensive expert consultations and enhance overall yield and tea quality. Along with a broader AI-powered decision-making platform, this module represents a significant advancement in equipping Sri Lanka's smallholder tea farmers with modern and scalable agricultural technology.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research is to develop an AI-driven decision support system that empowers newcomers and smallholder tea farmers in Sri Lanka by simplifying cultivation practices and improving productivity. Recognizing that many smallholders and new cultivators lack access to advanced technology, expert knowledge, and affordable diagnostic tools, this system provides an accessible platform to support their needs. By offering easy-to-use guidance on disease detection, pest identification, resource optimization, and practical cultivation market strategies, the AI platform enables users to reduce risks, make informed decisions, and establish sustainable and profitable tea businesses from the outset.

2.2 Specific Objectives

There are several specific objectives that must be reached in order to achieve the overall objective described above.

Accurate Detection of Tea Leaf Diseases and Pests

- Develop and train deep learning models such as CNNs/R-CNNs using high-quality, locally relevant tea leaf image datasets from the Tea Research Institute (TRI) of Sri Lanka. These models will accurately identify multiple diseases (like blister blight, red rust, brown blight etc.) and pests (such as shot hole borer).

User-Friendly Image Upload and Feedback Platform

- Design a simple, accessible application where smallholder farmers and newcomers can upload photos of affected tea leaves and receive fast, understandable results. The system will provide the name of the disease or pest, a clear description, and the severity level of

the infestation.

Treatment Guidance with Eco-Friendly Recommendations

- Integrate an expert treatment guidance module within the platform that suggests practical, environment-friendly treatment methods specific to the diagnosed problem, encouraging sustainable pest and disease management practices.

Validation with Real-World Data and Expert Collaboration

- Test, validate, and refine the system's accuracy and usability using real-world TRI datasets and feedback from Tea Research Institute experts, ensuring the model's effectiveness in practical Sri Lankan tea plantation conditions.

Detection and Management of Invasive Plant Species

- Extend the system's capabilities to identify and monitor invasive plant species such as *Lantana camara*, *Tridax procumbens*, and *Vernonia cinerea* that threaten tea plantations. Incorporate AI-based detection and control recommendations to support timely intervention, reduce competition for resources, and promote sustainable tea cultivation.

3. METHODOLOGY

3.1 Project Overview

The proposed system is designed to provide a comprehensive AI-powered solution for Sri Lanka's tea industry, addressing critical challenges such as declining yields, quality inconsistencies, labor shortages, and market inefficiencies. The system integrates four core modules: automated tea leaf grading, disease detection, yield prediction, and market demand forecasting into a unified decision-support platform accessible to tea estates, factories, and smallholder farmers. Figure 3.1 illustrates the overall system architecture.

Among these modules, the tea leaf disease and pest detection component is the primary focus of this project. It employs deep learning techniques, particularly Convolutional Neural Networks (CNN), to accurately detect major tea leaf diseases such as blister blight, red rust, white spot, algal leaf spot, red leaf spot, anthracnose, and pink disease, as well as pest infestations including tea mosquito bug, red spider mite, tea tortrix, and shot hole borer. By analyzing high-resolution images collected from tea estates and leveraging real-world local data from the Tea Research Institute (TRI) of Sri Lanka, the system ensures early and precise identification of affected leaves.

In addition to detection, the system provides treatment guidance, offering actionable suggestions based on expert knowledge to help farmers and estate managers take immediate corrective measures. A user-friendly dashboard consolidates disease alerts and treatment suggestions, enabling stakeholders to manage crop health effectively and proactively.

The expected outcome is a robust AI-powered decision support tool that empowers stakeholders to reduce crop loss, improve plant health, and optimize tea production. While the disease detection module is the centerpiece, the integration with grading, yield prediction, and market demand forecasting ensures a comprehensive platform that

enhances overall productivity and sustainability in the Sri Lankan tea industry.

3.2 System Architecture Diagram

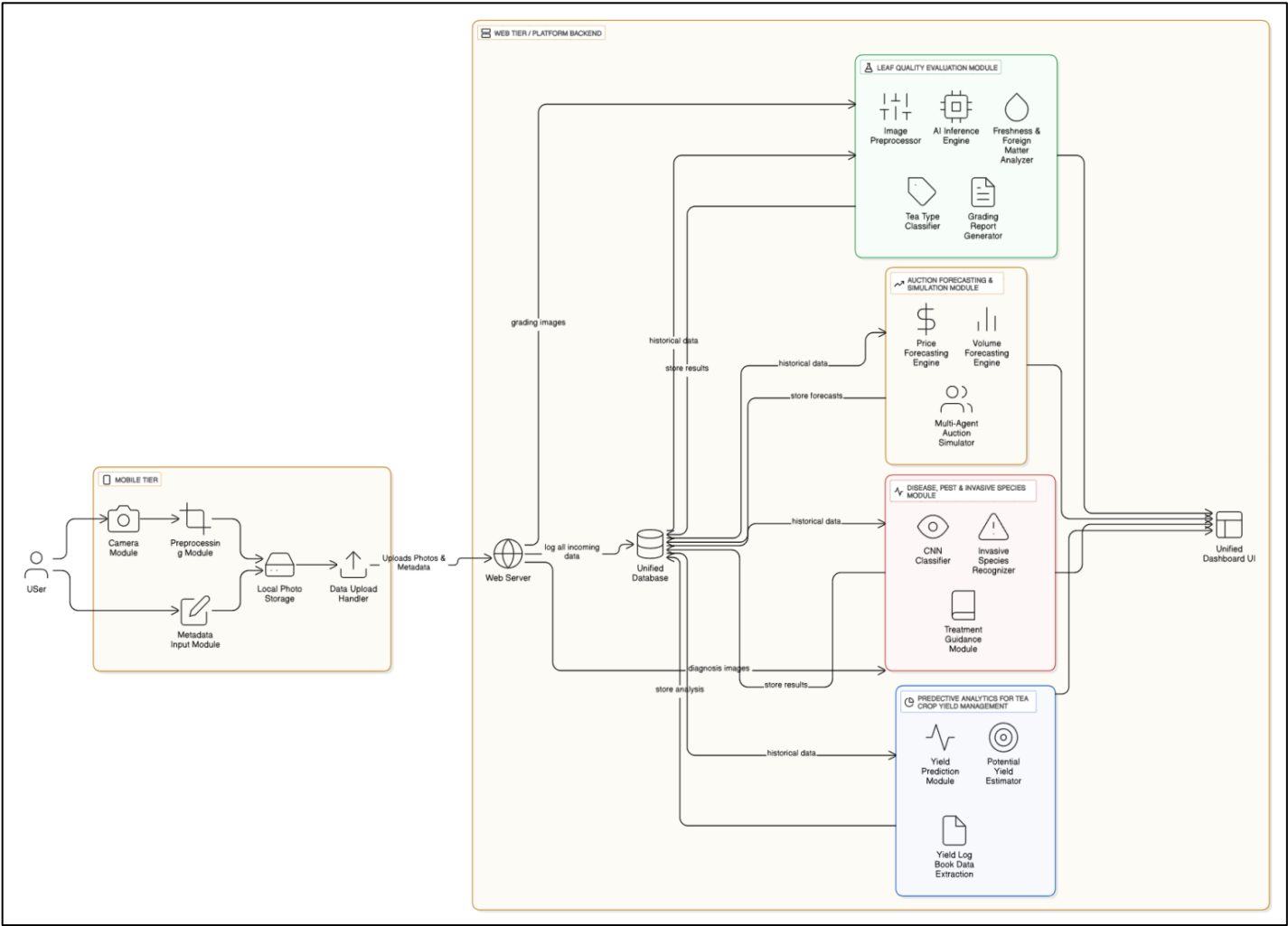


Figure 3. 1 : Overall System Architecture

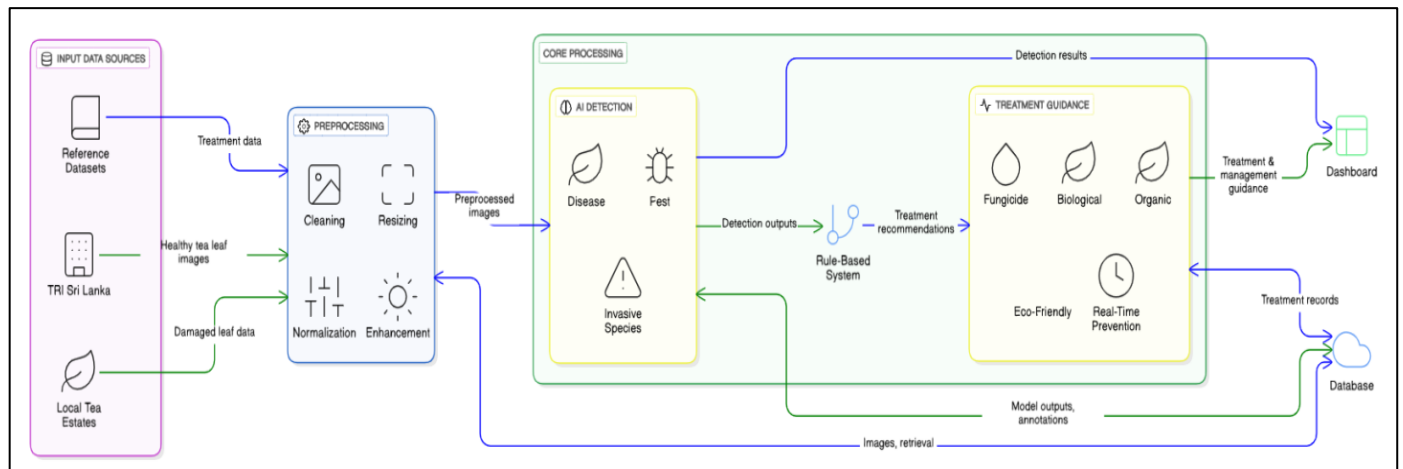


Figure 3. 2 : System Architecture of disease detecting component

3.3 System Overview

As shown in Figure 3.1, the system starts by collecting images and data from tea estates, factories, and industry experts. AI modules process these inputs to create useful insights, which are shared through a central dashboard with stakeholders.

The first component, Automated Tea Leaf Grading, uses image analysis to assess leaf size, color, and texture. This ensures consistent grading standards and lowers the need for manual inspection.

As shown in Figure 3.2, the second and main focus of this report, AI-Based Detection of Tea Leaf Diseases, Pest Infestations, Invasive Species, and Treatment Guidance, serves as the core decision-support feature. Tea leaf images first undergo preprocessing, which includes cleaning, resizing, normalization, and labeling. This is followed by augmentation to address real-world variations such as lighting and orientation. The processed images are then analyzed using Convolutional Neural Networks (CNNs) and deep learning classifiers to identify major diseases such as blister blight, red rust and brown blight, as well as pests like shot hole borer.

In addition to disease and pest detection, the system is also trained to recognize invasive plant species such as *Lantana camara*, *Tridax procumbens*, and *Vernonia cinerea*. These

species compete with tea bushes for nutrients, water, and sunlight, reducing yield and increasing management costs. By integrating invasive species recognition into the same AI framework, the system supports proactive monitoring and management, helping farmers maintain healthy plantations and prevent ecological imbalance.

Once detected the diseases, the Treatment Guidance module offers expert recommendations. These include fungicides, organic options, and eco friendly suggestions. This information is provided in real time through the dashboard for immediate action. The module is trained on annotated datasets from the Tea Research Institute (TRI) of Sri Lanka and local estates, making it relevant to regional conditions. Its performance improves through transfer learning and ensemble techniques, while cloud storage handles images, model outputs, and treatment records.

Another components, Predictive Analytics for Tea Crop Yield forecasts production using historical yield, climate, soil, and breeding data. Tea Market Demand Prediction uses time-series forecasting on auction data to match production with market needs.

Finally, the unified dashboard combines grading results, disease alerts with treatment guidance, yield forecasts, and market insights. Stakeholders including farmers, small factories, and new industry entrants receive combined alerts and recommendations to aid in data-driven decision-making.

By integrating image-based detection with treatment guidance and predictive analytics, the system offers a scalable and practical solution to improve tea crop health, reduce losses, and encourage sustainable cultivation.

Summary of technologies, techniques, architectures and algorithms used for the classification of Tea leaf, pest and invasive species detection.

Technologies	Python, TensorFlow, Keras
Techniques	Image preprocessing, Data augmentation, Feature extraction from images, Web based visualization, Labeling and annotation
Algorithms	CNN, RCNN, Transfer learning, Deep learning classifiers, Classification models for leaf and pest diseases
Architectures	Client-server architecture for dashboard

Table 3.1 : Technologies, techniques, architectures and algorithm

3.4 Software Development Process

The proposed AI-driven tea leaf disease and pest detection system with treatment guidance follows the Agile Software Development Lifecycle (SDLC). The Agile methodology was chosen for its flexibility and iterative nature, which allows the system to be refined continuously based on stakeholder feedback and real-world conditions.

To manage development effectively, the project adopts the SCRUM framework, ensuring collaboration through regular sprint planning, progress reviews, and backlog updates. This approach helps the team remain adaptable to challenges such as dataset variability, model accuracy, and field-level deployment needs. Overall, it ensures the system is both technically robust and practically applicable in real agricultural environments.



Figure 3.3 : Agile Methodology

3.4.1 Requirements Gathering and Analysis

To align the system with real-world needs, a comprehensive requirements gathering process was conducted through several approaches:

- **Field Visits and Expert Discussions:**

Multiple visits were made to the Hanthana Tea Research Institute (TRI), where discussions with the Head of TRI provided deep insights into the challenges faced by the tea industry, especially the impact of leaf and pest diseases on yield and quality. In addition to this, several officers from different divisions at the TRI Thalawakelle branch were contacted to obtain further practical insights regarding disease detection and management in varying climatic and environmental conditions. Moreover, officers from the Sri Lanka Tea Board were also consulted to gather policy-level and operational perspectives related to disease monitoring and tea quality assurance. These consultations helped refine the scope of the system and highlighted the need for early detection and timely intervention.

- **Exploration of Historical and Practical Knowledge:**

A visit to the Ceylon Tea Museum enabled the collection of photographic material and historical information, offering a broader perspective on tea production and its long-standing challenges. This enhanced contextual understanding while providing valuable visual references for the project.

- **Literature and Book References:**

Several books and academic publications on tea diseases and pest management were reviewed. In addition, a visit was made to the Gannoruwa Agriculture Library, where several specialized agricultural references and tea pathology materials were consulted to deepen the understanding of disease patterns and management techniques. These resources identified the most prevalent diseases in Sri Lanka and existing

detection/management practices, strengthening the scientific foundation of the project.

By combining expert input, field observations, and academic study, a solid base for the requirements analysis was established, ensuring the system directly addresses industry-relevant problems while being scientifically grounded.

3.4.2 Feasibility Study

- **Technical Feasibility**

The project requires expertise in machine learning and computer vision, specifically for detecting leaf and pest diseases. Frameworks such as TensorFlow/PyTorch and OpenCV will be used to build and train detection models. The system will be developed as a web-based and/or mobile application, with the back-end designed to support scalable data processing. Databases like MongoDB or MySQL will be used to store and manage collected image datasets, ensuring efficient data handling and real-world applicability.

- **Economic Feasibility**

The system is designed to be cost-effective by leveraging cloud-based resources and open-source libraries. This reduces development expenses while maintaining scalability and efficiency. Low operational costs make the project financially sustainable and accessible to stakeholders in the tea industry.

- **Operational Feasibility**

The solution aims to be practical for researchers, tea estate managers, and farmers by providing an intuitive interface and clear treatment guidance. Observations and informal discussions during visits to Hanthana TRI and the Ceylon Tea Museum helped enhance understanding of real-world challenges in tea cultivation, shaping the system's design and functionality.

- **Scheduling Feasibility**

Development will follow an Agile methodology with iterative sprints, allowing flexibility and continuous progress tracking. Milestones are aligned with research deadlines, ensuring the system is delivered on time while maintaining high standards of performance and usability.

3.4.3 Dataset

The dataset for the proposed tea leaf disease and pest detection system will consist of high-resolution images of both healthy and affected leaves. Images will capture major diseases such as blister blight, red rust, and brown blight, as well as pest infestations like shot hole borer.

In addition, the dataset will include samples showing the effects of invasive pest species such as *lantana camara*, *tridax procumbens* and *vernonia cinerea* that have recently begun affecting Sri Lankan tea plantations. These species, which are not native to the region, pose significant threats to tea cultivation by spreading rapidly and being resistant to traditional control methods. Including such cases in the dataset ensures that the system is capable of identifying and differentiating between local and invasive species, thereby improving its adaptability and practical value in diverse field conditions.

Data will be collected from the Hanthana Tea Research Institute (TRI), Thalawakelle Tea Research Institute (TRI) and several tea estates, supplemented by publicly available datasets, and annotated with disease or pest labels. Preprocessing and augmentation techniques will be applied to enhance model robustness. Metadata such as lighting conditions, leaf characteristics, and environmental context will also be recorded to improve real-world applicability.

The dataset will be stored in a structured database (e.g., MySQL or MongoDB) for training and evaluation of the AI models.

3.4.4 Implementation

The implementation phase of the proposed AI-driven tea leaf disease and pest detection system focuses on transforming the design and requirements into a fully functional platform that delivers accurate detection and actionable treatment guidance. The key stages are as follows:

Model Development:

- CNNs and other suitable deep learning architectures will be trained on the curated dataset of tea leaf images.
- Preprocessing techniques such as resizing, normalization, and data augmentation will be applied to improve model robustness and generalization across diverse field conditions.

System Integration:

- The trained AI models will be integrated into a user-facing platform, allowing users to capture or upload images of tea leaves for analysis.
- The system will provide predictions for detected diseases and pest infestations, along with TRI-approved treatment suggestions for early intervention.

Database Management:

- Image data, annotations, and model outputs will be stored in a structured database (e.g., MySQL or MongoDB) to ensure efficient retrieval, scalability, and management.

Deployment Preparation:

- The system will be optimized for field deployment, considering variable lighting, leaf orientation, and connectivity limitations in tea estates.
- Optimization techniques will ensure fast and accurate predictions for near real-time usage.

User Interface:

- A simple and intuitive interface will be designed to enable farmers, field officers, and researchers to use the system with minimal technical knowledge.
- Users will be able to view detection results, access treatment guidance, and monitor disease trends effectively.

3.4.5 Testing

The testing phase ensures that the AI-driven tea leaf disease and pest detection system is accurate, reliable, and suitable for field use.

Unit Testing:

Individual modules, including image preprocessing, AI model predictions, and database operations, will be tested for correct functionality.

Integration Testing:

Ensures smooth communication between AI models, database, and user interface, confirming predictions and treatment guidance are accurately delivered.

Model Evaluation:

AI models will be assessed using metrics such as accuracy, precision, recall, and F1-score on a validation dataset, including diverse leaf images from different estates.

System Testing:

The complete system will be tested in simulated and field conditions to verify real-time detection, responsiveness, and usability.

User Acceptance Testing (UAT):

Farmers, field officers, and TRI researchers will evaluate the system for practicality, clarity of guidance, and ease of use, with feedback incorporated into final refinements.

3.4.6 Grant Chart



Figure 3.4 : Gantt chart

3.4.7 Work Breakdown Structure(WBS)

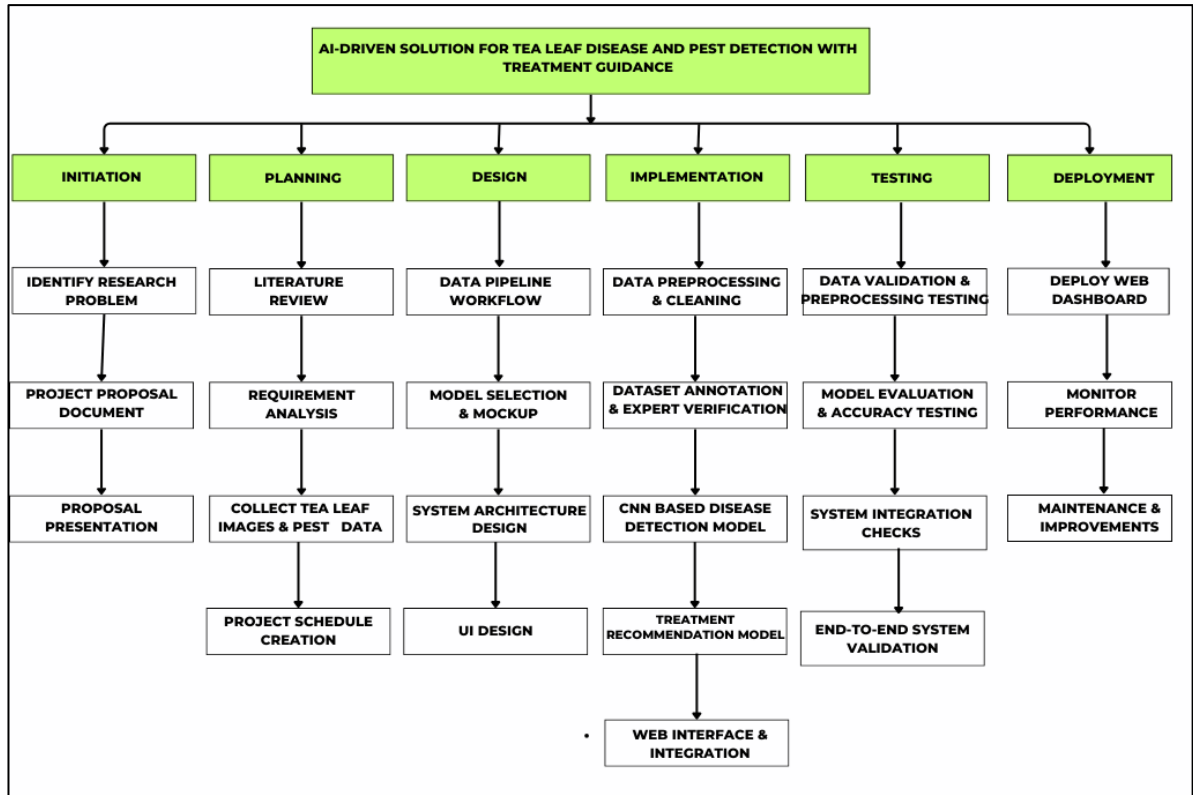


Figure 3.5: Work Breakdown Structure

4. PROJECT REQUIREMENTS

4.1 Functional requirements

1. Image Input and Preprocessing :

- The system should let users (farmers or researchers) upload or take photos of tea leaves.

2. Leaf Disease, Pest, and Invasive Species Detection:

- The system should detect whether the leaf is healthy or affected by any disease or pest.
- The system should identify the specific disease or pest (e.g., blister blight, shot hole borer etc.).
- The system should identify the specific invasive species (e.g., lantana camara, tridax procumbens etc.)

3. Treatment Guidance :

- The system should give treatment suggestions based on the detected disease or pest.
- Recommendations should include multiple approaches (chemical, biological, or eco-friendly methods etc.).
- The system should also suggest preventive and management strategies to reduce and stop the problem from happening again.

4. User Interaction and Guidance :

- The system should show results and treatment steps in a simple, clear interface with text and visuals.
- Visual indicators, icons, and descriptive text should help users easily understand the condition and recommended actions.

5. Performance and Accuracy :

- The system should be reliable, with at least 85% accuracy in detecting diseases, pests and invasive species.
- The results should be delivered quickly.

4.2 Non-functional requirements

1. Performance :

- The system should give detection results quickly (within 5 seconds per image).
- It should maintain high accuracy (at least 85%) when identifying diseases and pests.

2. Reliability :

- The system should work consistently without frequent crashes or errors.
- It should provide stable results even when internet connectivity is slow or unstable.

3. Usability :

- The interface should be simple and easy to use for farmers with minimal technical knowledge.
- Visuals and guidance should be clear, with simple language and step-by-step treatment advice.

4. Scalability :

- The system should handle a growing number of users without performance issues.
- It should allow adding new diseases, pests, or treatment methods in the future.

5. Security :

- User-uploaded images and reports should be stored safely without unauthorized access.
- The system should protect user data privacy.

6. Maintainability :

- The system should allow easy updates to the knowledge base with new agricultural research.
- Developers should be able to fix bugs or add new features without major rework.

4.3 Software requirements

- **Frontend Requirements**

- Web Access – The system should work smoothly on modern browsers (Chrome, Edge) for broad accessibility and a seamless user experience.

- **Backend & Development Requirements**

- Development Tools – Visual Studio Code and MySQL Workbench for backend development and database management.
- Programming Languages – Python for backend logic, AI/ML models, and APIs; HTML/CSS/JS for frontend.
- Frameworks & Libraries – Flask for APIs, TensorFlow, Pandas, NumPy, Matplotlib, and Scikit-learn for AI/ML and data processing.
- Version Control – Git and GitHub for collaboration and code management.

4.4 User requirements

- Users should receive instant identification of the detected disease or pest from the uploaded image.
- The system should display the confidence level of detection (e.g., percentage of accuracy).
- Users should be provided with clear, step-by-step treatment recommendations for each identified disease or pest.
- Users should be able to navigate the application without prior technical knowledge. Clear icons, instructions, and visuals should guide them through the process.

4.5 Expected Test Cases

Test case ID: SID_Test_01			Test designed by: Wijerathne H.M.K.G.D.P.		
Test title: Tea Leaf Disease and Pest Detection with Treatment Guidance			Test designed day: 2025/8/29		
Test priority (High/Medium/Low): High			Test executed by: Wijerathne H.M.K.G.D.P.		
Module name: Leaf Disease & Pest Detection			Test executed day: 2025/8/31		
Description: Detect diseases or pests from uploaded tea leaf images and display treatment guidance.					
Pre-conditions: User logged into system and image of tea leaf available.					
Testing steps: Login to the system as a registered user. Navigate to the “Leaf Disease Detection” section. Upload a clear tea leaf image (affected sample). 4. Click the “Detect Disease/Pest” button. 5. View detection result and treatment recommendations.					
Test ID	Test Inputs	Expected Output	Actual Output	Result (Pass/Fail)	Comments
SID_Test_01	Upload tea leaf image with visible blister blight symptoms.	System detectss blister blight and provides both organic & chemical treatments.	System detects blister blight and provides recommended treatments.	Pass	Detection Accuracy

Table 4. 1 : Expected Test Case1

Test case ID: SID_Test_02		Test designed by: Wijerathne H.M.K.G.D.P.			
Test title: Pest Infestation Detection – Shot Hole Borer		Test designed day: 2025/8/29			
Test priority (High/Medium/Low): High		Test executed by: Wijerathne H.M.K.G.D.P.			
Module name: Leaf Disease & Pest Detection		Test executed day: 2025/8/31			
Description: Detect pest infestation from uploaded tea leaf images and display treatment guidance.					
Pre-conditions: User logged into system and image of tea leaf with pest damage available.					
Testing steps: 1. Login to the system as a registered user. 2. Navigate to the “Leaf Disease Detection” section. 3. Upload a tea leaf image showing shot hole borer damage. 4. Click the “Detect Disease/Pest” button. 5. View detection result and treatment recommendations.					
Test ID	Test Inputs	Expected Output	Actual Output	Result (Pass/Fail)	Comments
SID_Test_02	Upload tea leaf image showing damage caused by shot hole borer	System detects shot hole borer infestation and provides treatment guidance (organic & chemical options).	System detects shot hole borer and provides recommended treatment options.	Pass	Detection Accuracy

Table 4. 2 : Expected Test Case2

5. COMMERCIALIZATION

The commercialization strategy of the Tea Leaf and Pest Disease Detection with Treatment Guidance and Invasive Species Detection System focuses on supporting smallholder growers, plantation managers, and extension officers with affordable, AI-powered tools for early detection and treatment. Delivered as a mobile application with subscription plans, the system reduces crop loss, improves tea quality, and minimizes pesticide misuse.

Partnerships with the Sri Lanka Tea Board, Colombo Tea Traders' Association, and Hantana Tea Research Institute will build trust and promote adoption, while collaborations with agrochemical and organic solution providers will enhance treatment recommendations. Marketing will leverage industry forums, cooperatives, and digital platforms to reach both small and large-scale users.

In the long term, the system aims to become a core decision-support tool in Sri Lanka's tea sector and expand regionally, ensuring sustainability through continuous user feedback and field validation.

6. BUDGET AND BUDGET JUSTIFICATION

The estimated budget mainly covers travel expenses for data collection and website hosting. Traveling costs are required for visits to the Tea Research Institute (TRI) of Sri Lanka, the Sri Lanka Tea Board, and selected tea estates to collect image datasets, consult experts, and validate the AI model. A portion of the budget is also allocated for hosting the AI-based detection and treatment guidance system online, ensuring easy access for farmers and stakeholders. As the system uses open-source tools and local resources, no major expenses are expected for software or cloud services.

The below table 6.1 depicts the overall budget of the proposed system.

Expenses	
Requirement	Cost (Rs.)
Travelling cost for data collection	8,000.00
Cost of Hosting	4,990.00
Total Cost	12,990.00

Table 6. 1 : Expenses for the proposed system

REFERENCES

- [1] E. Obadaarachchi, D. Saputhanthri, P. Vithanarachchi, Y. Zuhry, and R. Rathnayake, "GreenGuard: Mobile Application for Real-Time Tea Leaves Disease Detection and Treatment Guidance," in *Proc. 2024 6th International Conference on Advancements in Computing (ICAC)*, Colombo, Sri Lanka, Dec. 12-13, 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10851118>
- [2] S. Sharath, P. Jayanthi, K. Nirmaladevi, N. Sri Sanjai, and A. Thamiz Selvan, "Enhanced Tea Leaf Disease Detection using Deep Learning," in *Proc. 2024 Second International Conference on Emerging Trends in Information Technology and Engineering (ICETITE)*, Vellore, India, Feb. 22-23, 2024. [Online]. Available: <https://doi.org/10.1109/ic-ETITE58242.2024.10493308>
- [3] G. Hu and M. Fang, "Using a Multi-Convolutional Neural Network to Automatically Identify Small-Sample Tea Leaf Diseases," *Sustainable Computing: Informatics and Systems*, vol. 35, p. 100696, Sep. 2022. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S2210537922000385>
- [4] S. Bhowmik, A. K. Talukdar, and K. K. Sarma, "Detection of Disease in Tea Leaves Using Convolution Neural Network," in *Proc. 2020 Advanced Communication Technologies and Signal Processing (ACTS)*, Silchar, India, Dec. 4-6, 2020. [Online]. Available: <https://doi.org/10.1109/ACTS49415.2020.9350413>
- [5] G. Kaur, N. Sharma, and R. Gupta, "Tea Leaf Disease Detection Using Deep Learning Convolutional Neural Network Model," in *Proc. 2023 Int. Conf. on Evolutionary Algorithms and Soft Computing Techniques (EASCT)*, Bengaluru, India, Oct. 20-21, 2023. [Online]. Available: <https://doi.org/10.1109/EASCT59475.2023.10393773>

- [6] R. Singh, N. Sharma, and R. Gupta, "Proposed CNN Model for Tea Leaf Disease Classification," in *Proc. 2023 2nd Int. Conf. on Applied Artificial Intelligence and Computing (ICAAIC)*, Salem, India, May 4-6, 2023. [Online]. Available: <https://doi.org/10.1109/ICAAIC56838.2023.10140680>
- [7] R. D. Praba, K. Kavitha, S. Abinaya, P. Abarna, K. S. Shri, and S. N. Shivappriya, "Automatic Tealeaf Disease Detection Using Machine and Deep Learning Method," in *Proc. 2023 2nd Int. Conf. on Advancements in Electrical, Electronics, Communication, Computing and Automation (ICAECA)*, Coimbatore, India, Jun. 16-17, 2023. [Online]. Available: <https://doi.org/10.1109/ICAECA56562.2023.10200312>
- [8] S. Vats, V. Kukreja, and S. Mehta, "Tea Leaf Disease Detection: Federated Learning CNN Used for Accurate Severity Analysis," in *Proc. 2024 IEEE Int. Conf. on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)*, Gwalior, India, Mar. 14-16, 2024. [Online]. Available: <https://doi.org/10.1109/IATMSI60426.2024.10503207>
- [9] R. Thangaraj, B. Vadivel, T. Aravind, L. Chandhru, K. Harish, and S. Eswaran, "Tea Leaf Disease Identification using Improved Convolution Neural Network," in *Proc. 2023 Int. Conf. on Digital Applications, Transformation & Economy (ICDATE)*, Miri, Sarawak, Malaysia, Jul. 14-16, 2023. [Online]. Available: <https://doi.org/10.1109/ICDATE58146.2023.10248508>
- [10] A. Saini, K. Guleria, and S. Sharma, "An Efficient Fine-Tuned Convolutional Neural Network Model for Tea Leaf Disease Classification," in *Proc. 2023 3rd Int. Conf. on Smart Generation Computing, Communication and Networking (SMART GENCON)*, Bangalore, India, Dec. 29-31, 2023. [Online]. Available: <https://doi.org/10.1109/SMARTGENCON60755.2023.10442044>
- [11] J. Sivalingam, L. N. Sivachandrabose, M. Loganathan, J. Sivakumaran, and R. Panchendrarajan, "tAssessee: Automatically Assessing Quality of Tea Leaves using

Image Processing Techniques,” in *Proc. 2022 22nd Int. Conf. on Advances in ICT for Emerging Regions (ICTer)*, Colombo, Sri Lanka, Nov. 30 - Dec. 1, 2022. [Online]. Available: <https://doi.org/10.1109/ICTer58063.2022.100240>

[12] C. Liu, J. Cai, Z. Wu, and J. Yang, “Tea Pest and Disease Detection Algorithm Based on Deep Learning,” in *Proc. 2025 5th Int. Conf. on Artificial Intelligence and Industrial Technology Applications (AIITA)*, Xi'an, China, Mar. 28-30, 2025. [Online]. Available: <https://doi.org/10.1109/AIITA65135.2025.11047622>

[13] C. Kariyawasam, L. Kumar, and S. Ratnayake, “Invasive Plant Species Establishment and Range Dynamics in Sri Lanka under Climate Change,” *Entropy*, vol. 21, no. 6, p. 574, Jun. 2019. [Online]. Available: <https://doi.org/10.3390/e21060574>

[14] K. G. Prematilake, “Weed Management in Tea – Recent Developments,” *Tropical Agricultural Research & Extension*, vol. 6, 2024. [Online]. Available: <https://tare.sljol.info/articles/5447/files/submission/proof/5447-1-19180-1-10-20200923.pdf>

[15] K. C. Kumhar, A. Babu, and S. N. Nisha, “Management of Tea (*Camellia Sinensis*) Diseases with Application of Microbes: A Review,” *Int. J. Agric. Sci.*, vol. 10, no. 2, Apr. 2022. [Online]. Available: <https://www.researchgate.net/publication/360262694>

[16] R. R. Kumar, R. Bhandari, A. K. Jain, N. Sharma, V. Sharma, and S. Chawla, “Hybrid CNN & Random Forest Model for Effective Tea Leaf Disease Diagnosis,” in *Proc. 2024 4th Int. Conf. on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)*, Bhilai, India, Jan. 11-12, 2024. [Online]. Available: <https://doi.org/10.1109/ICAECT60202.2024.10469122>

[17] Y. Li, Y. Lu, H. Liu, J. Bai, C. Yang, H. Yuan, X. Li, and Q. Xiao, “Tea leaf disease and insect identification based on improved MobileNetV3,” *Front. Plant Sci.*,

vol. 15, p. 1459292, Sep. 2024. [Online]. Available:
<https://doi.org/10.3389/fpls.2024.1459292>

[18] G. Vijayakanthan, R. Vaishali, and V. Abolghasemi, "Detection of Tea Leaf Diseases Using Deep Transfer Learning," in *Proc. 2024 Moratuwa Engineering Research Conference (MERCon)*, Moratuwa, Sri Lanka, Aug. 8-10, 2024. [Online]. Available: <https://doi.org/10.1109/MERCon63886.2024.10688521>